

Edge Analytics Enabled Acoustic Rain Gauge

Sajil C K
Research Assistant
ICFOSS

Thiruvananthapuram, India
sajilck@icfoss.org

Shafeek P M
Technical Coordinator
ICFOSS

Thiruvananthapuram, India
shafeek@icfoss.in

Manu Mohan M S
Research Assistant
ICFOSS

Thiruvananthapuram, India
manumohan@icfoss.org

Gopika T G
Research Associate
ICFOSS

Thiruvananthapuram, India
gopikatg@icfoss.org

Aiswarya Babu
FOSS Innovation Fellow
ICFOSS

Thiruvananthapuram, India
aiswaryababu@icfoss.org

Harikrishnan K P
Apprentice
ICFOSS

Thiruvananthapuram, India
harikrishnankp@icfoss.org

Abstract—Precise, low-cost, and low-power precipitation measuring devices are vital to weather forecasting. In this work, we demonstrate a non-mechanical acoustic rain gauge which can estimate rainfall with reasonable accuracy. The proposed system leverages low-power Internet of Things (IoT) devices coupled with deep learning algorithms to accomplish this objective.

Index Terms—audio signal processing, IoT, deep learning

I. INTRODUCTION

Environment stability and protection is key to continuation of life on Earth. Problems associated with environment pollution and global warming need to be addressed. One of the preliminary steps to achieve this is continuous monitoring of environmental parameters such as rain, wind, temperature, pressure, etc.

Rainfall is an important part of climate. Rainfall monitoring devices can be broadly classified into mechanical or non-mechanical. It is desirable to have accurate monitoring devices which are automatic and non-mechanical. The manual methods are expensive and require human intervention. Mechanical rain gauges often use a tipping bucket mechanism where each tip is associated with a minimum amount of rainfall it can detect. During heavy and high intensity rainfall, tipping bucket rain gauges are not accurate. Also they are affected by clogging, wind, temperature, humidity leading to measurement errors [1]. Frequent maintenance at remote locations is another concern.

Kerala state in India faced flood in 2018-2019 year which is attributed to heavy rain that happened in a short time. A network of low-cost and long-durable rain gauges can be used to create early flood warning systems.

Acoustic rain gauges do not use moving parts addressing many of the problems discussed with mechanical ones. The sound produced by rain drops varies depending on the rain drop size and impact which is associated with the amount of

rainfall. An enclosure material such as aluminum or stainless steel produces better sound with respect to rain intensity [2]. The sound data in a given time interval can be paired with actual rainfall readings to do function approximation using deep learning.

Micro-climate [3] modeling requires a wide array of sensor network for high resolution and frequent data collection. For such massive deployment, it is important to have rain gauges which are low-cost, low-power and having long standby time. In this work, we demonstrate an acoustic rain gauge which makes use of low-power Internet of Things (IoT) edge devices and deep learning algorithms. The proposed low-cost acoustic rain gauge can be mass produced and deployed for the purpose of micro-climate monitoring. Applications of the proposed rain gauge span various sectors and are integral to achieving Sustainable Development Goals (SDG) [4].

II. RELATED WORK

In the present climatic conditions, the significance of micro-climate monitoring systems is on the rise. Weather monitoring and prediction emerge as crucial elements in these systems. Rainfall measurement employs various methods, with manual and automated approaches being chosen based on the specific application and resource availability.

Currently, different types of rainfall measurement methods are used. Mechanical and Non-mechanical rain gauges are used to know how much rain has fallen. Siphon rain gauge and tipping bucket rain gauges are of mechanical rain gauge type. Both these rain gauges are meteorological instruments used to measure precipitation. The former utilises a bucket that tips after a specific rainfall amount. Due to the high rainfall intensity the tipping bucket rain gauge may read inaccurate measurements. There is a chance of data loss as it takes time to fill the tipping bucket when there is light rain. Also, if not properly maintained, it may provide inaccurate readings.

The Siphon rain gauge consists of a rain collector, a float chamber, an automatic recording clock, and a protective casing. The rain water flows through the float chamber and it collects into the rain collector. When the rainwater increases the automatic recording clock rotates and it plots the measurement of rainfall [5]. It is not suitable for automatic reading, typically requiring manual reading. The classification of different types of rainfall measurement methods are shown in Figure 1.

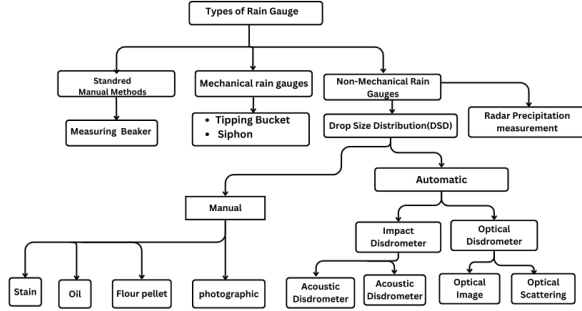


Fig. 1. Classification of rain gauges [6]

The Acoustic rain gauges are comparatively low cost. Deployment and maintenance of these types of rain gauges are easy. Researchers previously approached the acoustic rain gauge problem from a classification point of view [7]. Here we define the problem from a deep learning regression point of view.

III. METHODS

A. Experiment Setup

The block diagram 2 shows the overall pipeline used for data collection and inference. Figure 3 (a) and Figure 3 (b)

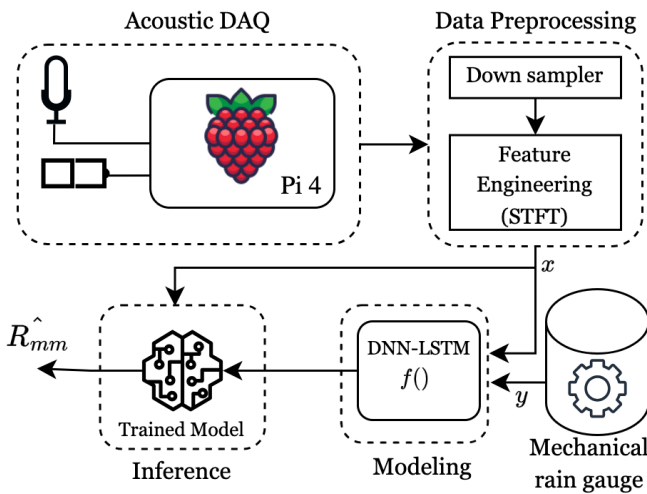


Fig. 2. Block diagram of Acoustic Rain Gauge

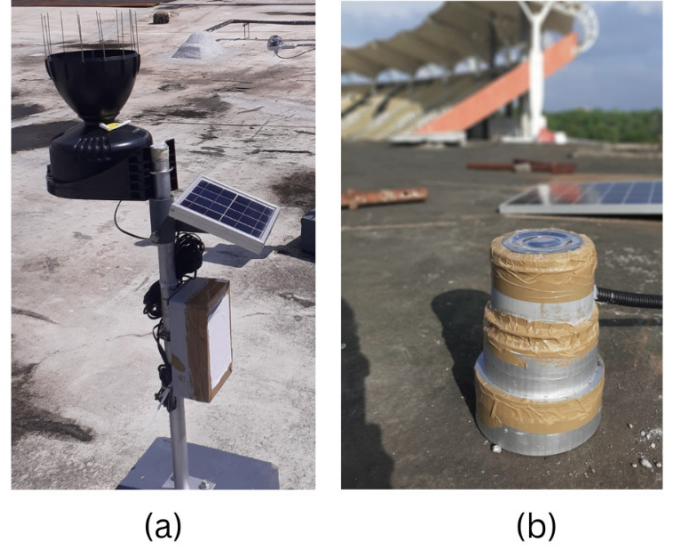


Fig. 3. Photographs of mechanical (a) and acoustic (b) rain gauge setups

show the traditional tipping bucket rain gauge module and the enclosure of our acoustic rain gauge respectively. These two setups facilitate real-time data collection and further development of a deep learning model for rainfall estimation.

Tipping bucket rain gauge (Davis® 0.2 mm Rain Gauge Smart Sensor) is tested and integrated with Arduino pro mini development board for data collection. It is configured in such a way that rain data collected will be sent to a Chirpstack server in every 3 minute interval using LoRaWAN technology. This setup is powered by a 3.8 battery with solar charger.

B. Sound Data Collection

The internal components of our acoustic rain gauge consists of Raspberry Pi 4 and a USB microphone connected to it. In-order to power up this setup we used a 50 Ah Li-ion battery pack with a solar charging system. A Python script is written to save wav files of given sampling rate, resolution, and duration along with timestamps as file names as soon as the Raspberry Pi boots up.

The rainfall audio data was collected across multiple days during the rainy season (November-December 2023) at the premise of Greenfield International Stadium, Thiruvananthapuram. A rainfall of 1mm to 8mm was observed during the period. The total time duration of recordings is more than 12 hours. The equivalent readings from mechanical rain gauge every 3 minutes is also provided.

The data from the USB mic contains audio files saved in wav format with a fixed duration. The parameters of the audio files (e.g. sampling rate, sample duration, bit size, total recording time, etc.) can be set in the 'config.yaml' file. The recorded wav files are saved with a timestamp format

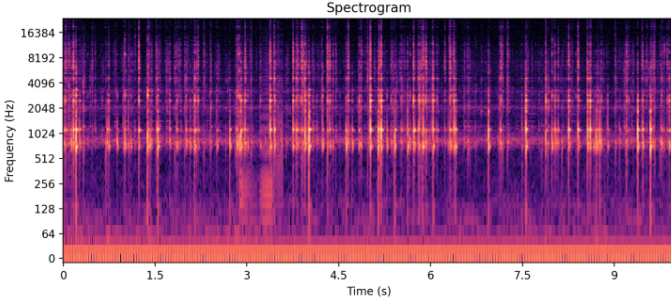


Fig. 4. Spectrogram of sample audio during rainfall. The vertical lines corresponds to rain drop impact sound on the enclosure

yyyy_mm_dd_hh_mm_ss_millisecond.wav. Figure 4 shows spectrogram of a sample audio file during rainfall.

The raw audio data collected along with readings from mechanical rain gauge is made available online at <https://www.kaggle.com/datasets/sajilck/rain-data-master-2023>. The recorded audio files are further processed for deep learning modeling.

C. Data Preprocessing

A preprocessing script is used to convert all audio files to a sample rate of 8000 samples/sec. The down-sampled version of the data set is also made available <https://www.kaggle.com/datasets/sajilck/rain-data-master-8k> via Kaggle. To create paired sequential input and associated target values, a data preparation [Kaggle script](#) is written. The output of this data preparation Kaggle script is made available as an input data set <https://www.kaggle.com/datasets/sajilck/rain-seq-master> for further deep learning modeling.

D. Feature Engineering

The Magnitude of Short Time Fourier Transform (STFT) of the audio signal was used as raw features for the LSTM input. The Python library Librosa [8] is used for handling the wav files and generating the STFT features.

E. Deep Learning Model for Regression

The audio data is a time series in nature. The system is designed to estimate rainfall every 3 minutes. Given the problem definition, sequential learning blocks such as Long Short-Term Memory (LSTM) are used as part of the Deep Neural Network (DNN).

The whole data set was split into train (75%) and test (25%) sets. The deep learning library Keras was used to create the deep neural network. The Mean Absolute Percentage Error (MAPE) of the trained DNN was evaluated on the test set and reported. The code for the data modeling is made public via a data modeling [Kaggle script](#).

F. Performance Metrics

To evaluate the performance of our rainfall estimation model, the following performance metrics were chosen.

- Mean Absolute Percentage Error (MAPE): The MAPE shows the comparison of predicted values (R_{mm}) with respect to actual values (R_{mm}) in percentages. The lower the value of MAPE, the better the performance of the model. The equation to compute MAPE is as follows.

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{R_{mm} - \hat{R}_{mm}}{R_{mm}} \right| \times 100 \quad (1)$$

- Coefficient of determination (R^2): The coefficient of determination shows us how good our regression model is compared to a baseline model that simply predicts the average value of R_{mm} . The possible values of R^2 ranges from 0 (worst) to 1 (best).

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \quad (2)$$

where SS_{res} is sum of squares of residuals and SS_{tot} is total sum of squares

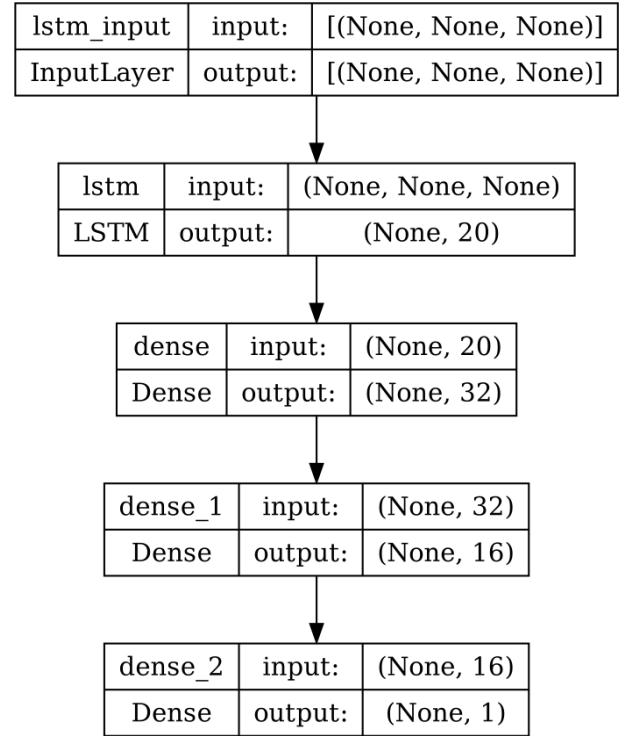


Fig. 5. Architecture of deep learning regression model used

IV. RESULTS AND DISCUSSION

A. Results of Regression Model

We tried a sequential deep neural network that can be trained to estimate rainfall using the mechanical rain gauge

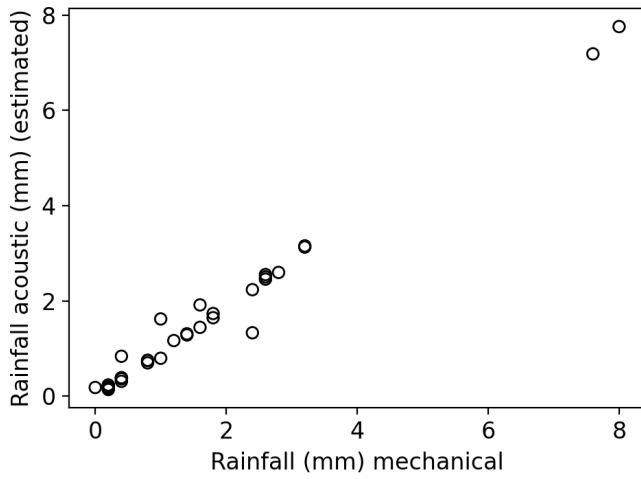


Fig. 6. Scatter plot of rainfall mechanical (mm) vs rainfall acoustic estimated (mm)

values as target.

Figure 6 shows a linear relationship between the rainfall recorded in mechanical rain gauge and our acoustic rain gauge. Table I shows Mean Absolute Percentage Error (MAPE), R^2 score and correlation coefficient (Spearman's) of DNN-LSTM model on test data.

TABLE I
PERFORMANCE OF LSTM MODEL

Epochs	Model	MAPE	R2	ρ (Spearman's)
25	DNN-LSTM	31.15%	0.9149	0.9597

It is evident from Table I that the results from our acoustic rain gauge matches with the readings from the mechanical rain gauge with an accuracy of $\sim 70\%$ and R^2 score of 0.9149.

V. CONCLUSION

We have demonstrated a non-mechanical acoustic rain gauge using IoT edge devices and deep learning algorithms. The proposed regression model system has a MAPE of 31.15%. The code and detailed experiment setup are made public in the GitHub repository https://github.com/cksajil/rainfall_monitor in the spirit of reproducible research. Future work involves reducing the percentage error of the acoustic rain gauge. The implementation of the idea into a lower specification hardware (e.g. Arduino Nano BLE Sense) using TinyML [9] framework can also be explored to improve power consumption and standalone deployment life of the device.

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